

Today's Outline

- 3D Scene Understanding

The goal of computer vision

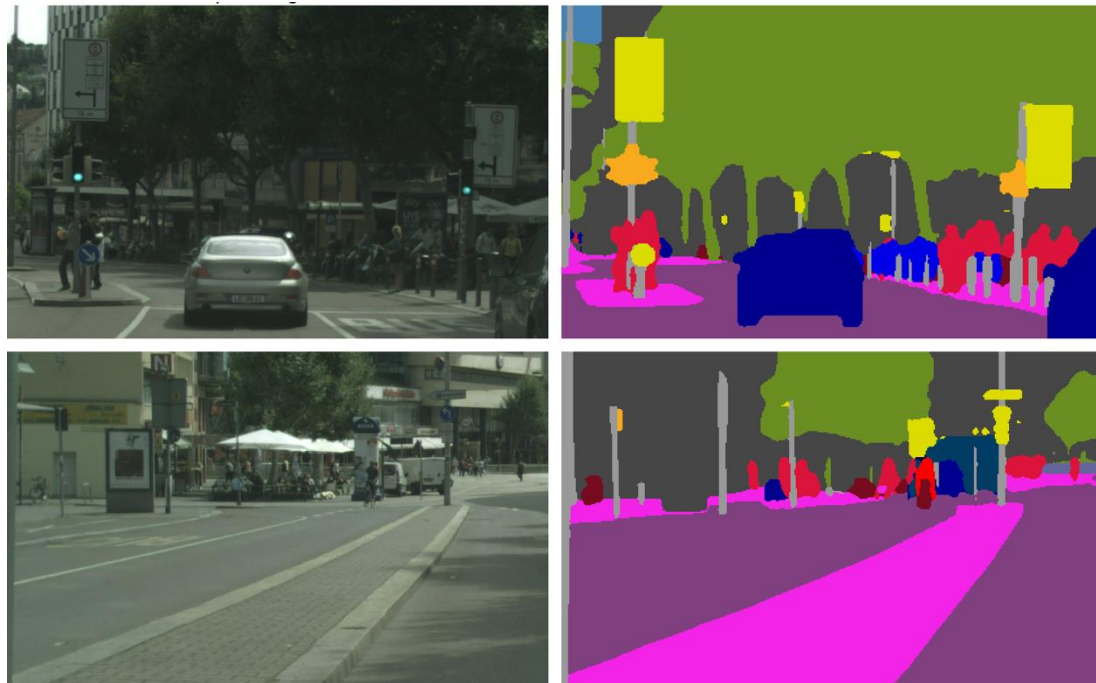
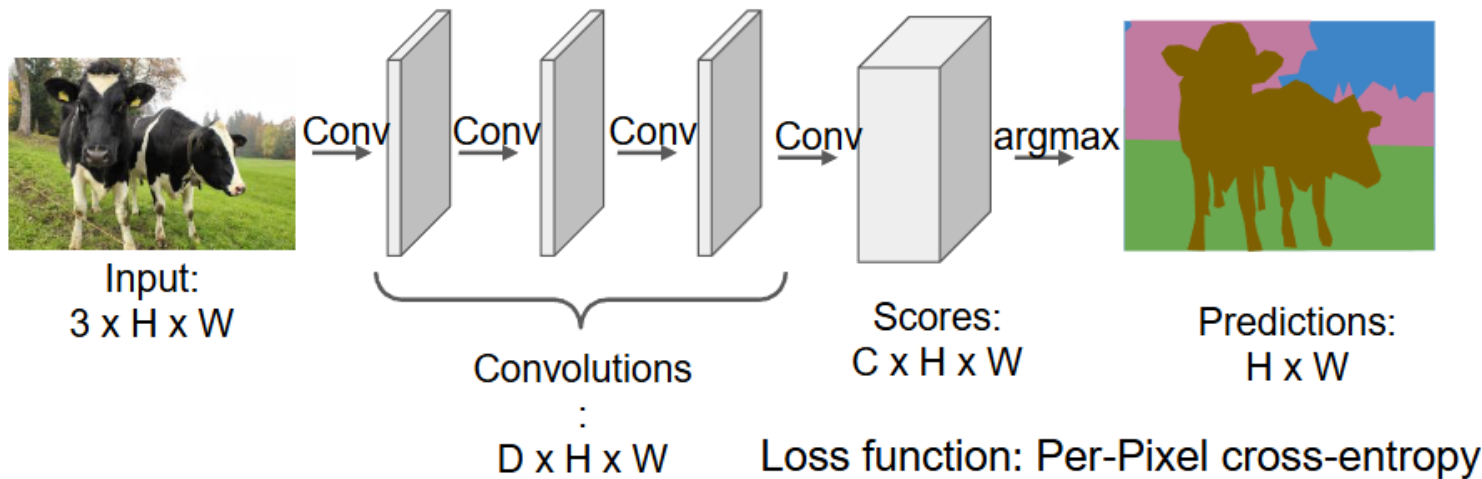


Image segmentation

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!

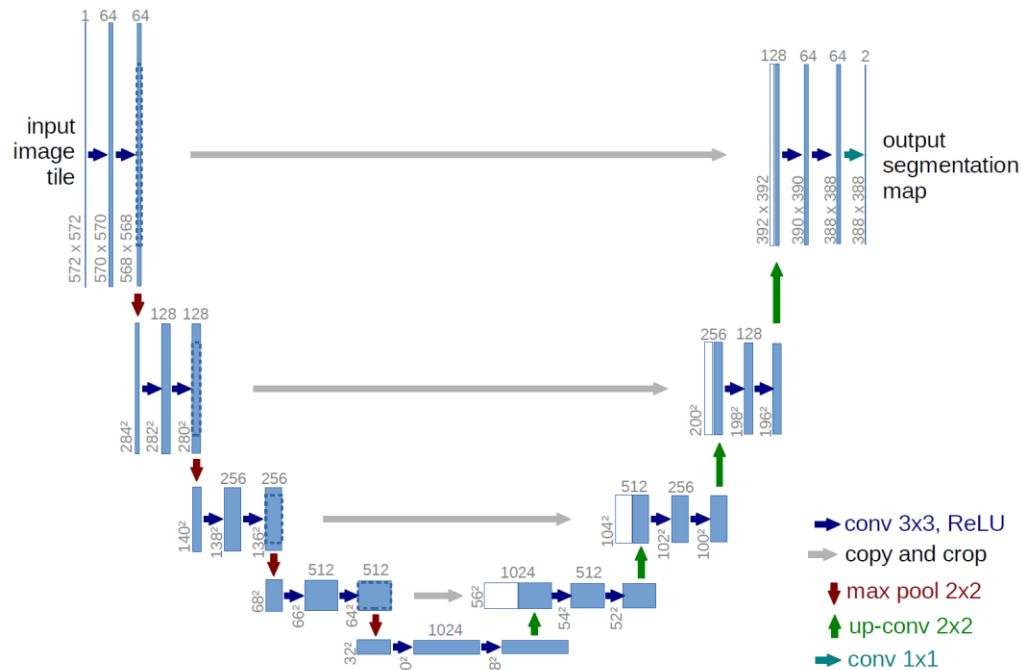


Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

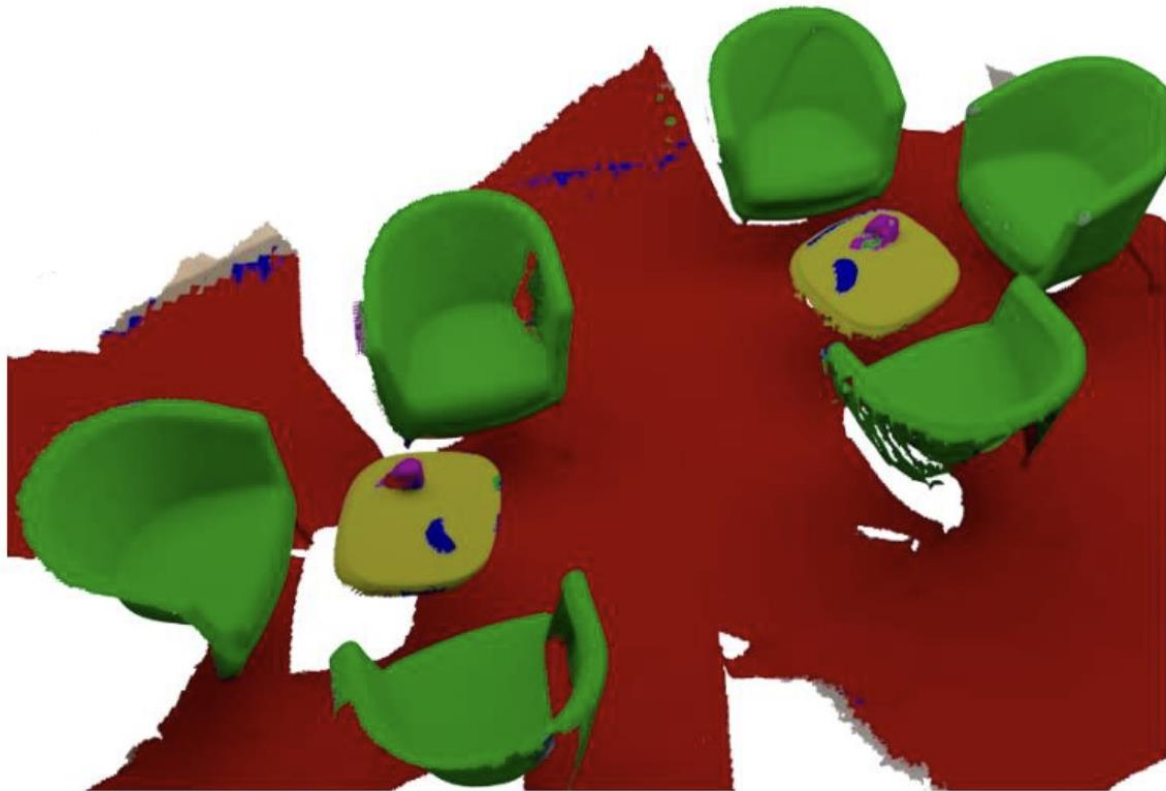
To Enrich the Spatial Perception 2D Image Understanding

- UNet

- ✓ Good speed
- ✓ Good accuracy
- ✓ Pyramid perception



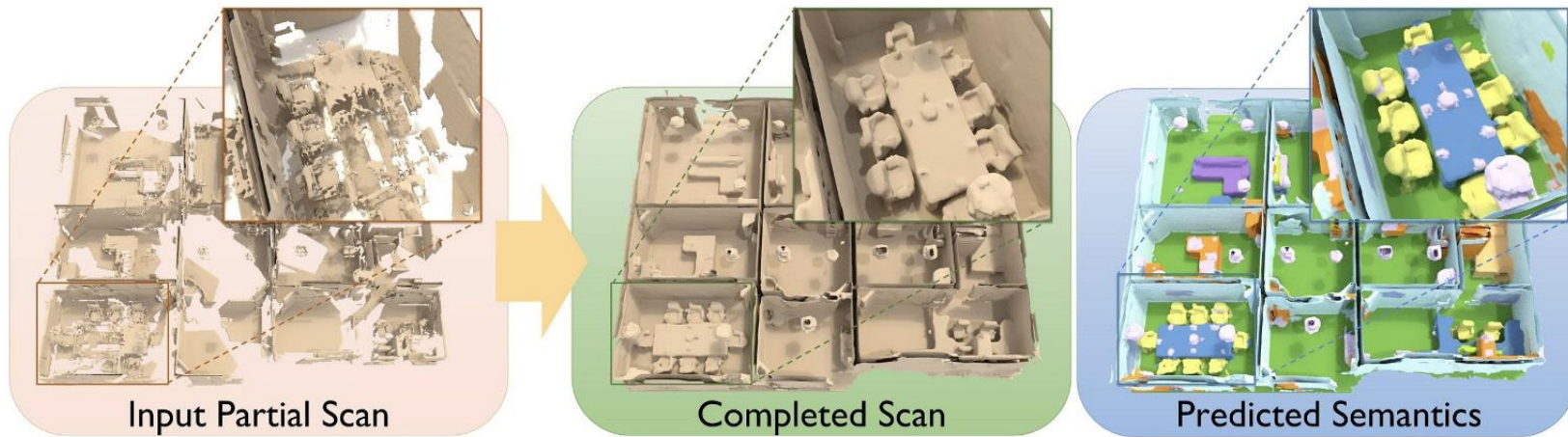
Scene understanding



Datasets for Indoor 3D Scenes: Scannet



3D Shape Completion: ScanComplete



Larger Scale 3D Datasets

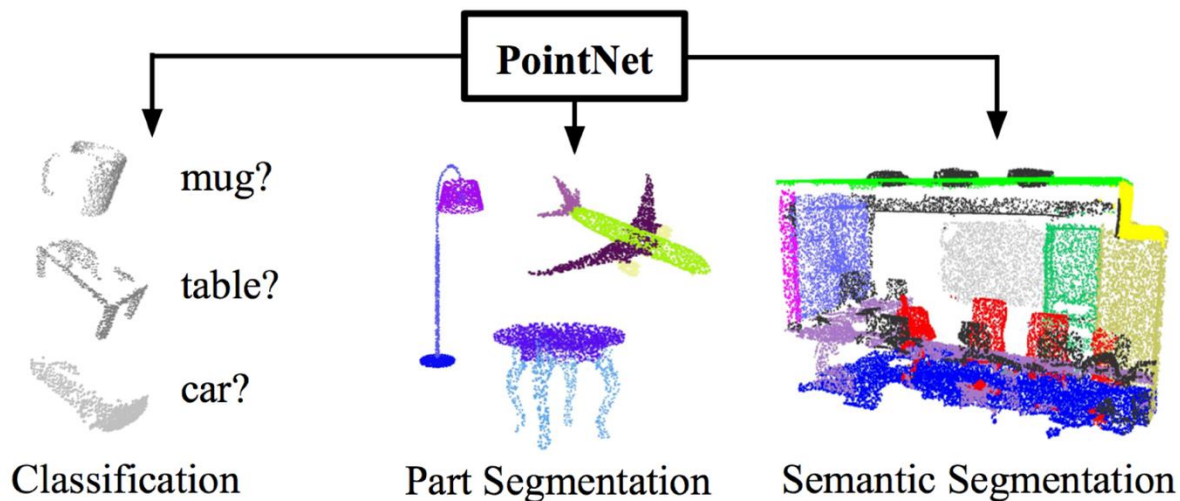
- ScanNet++:
- 1006 high-fidelity Scenes
- 1mm laser scans-DSLR images
- iPhone RGB-D-semantics



3D Point Cloud Segmentation

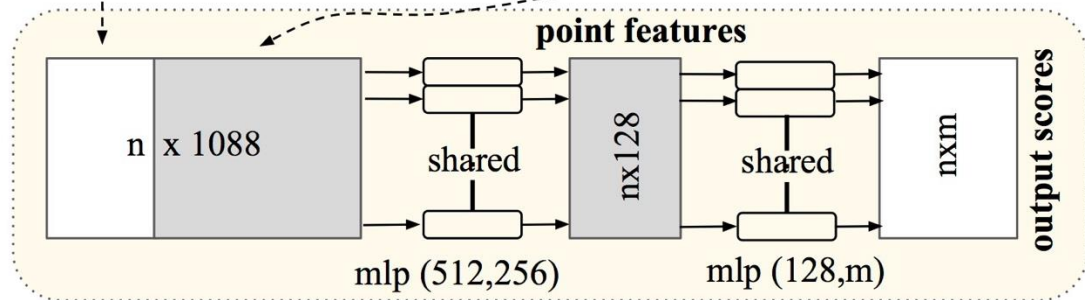
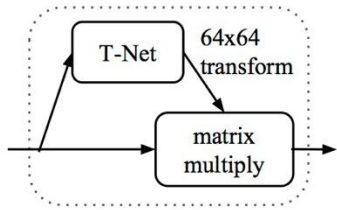
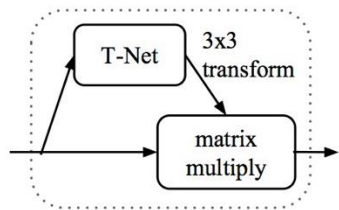
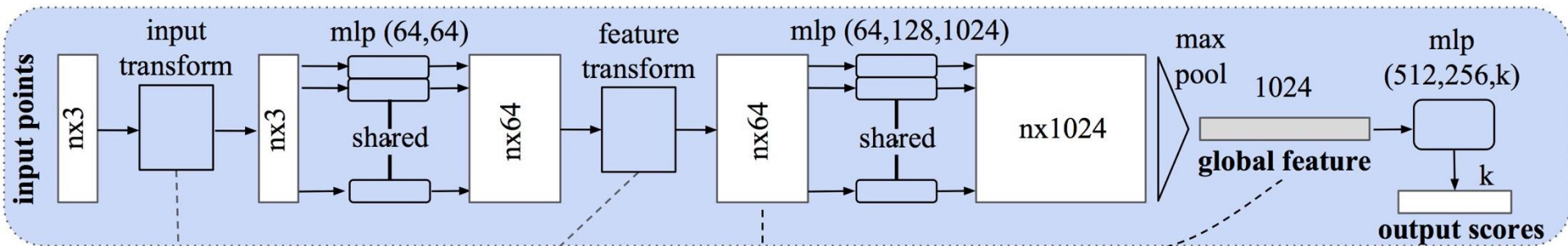
End-to-end learning for **scattered, unordered** point data

Unified framework for various tasks



PointNet

Classification Network



Segmentation Network

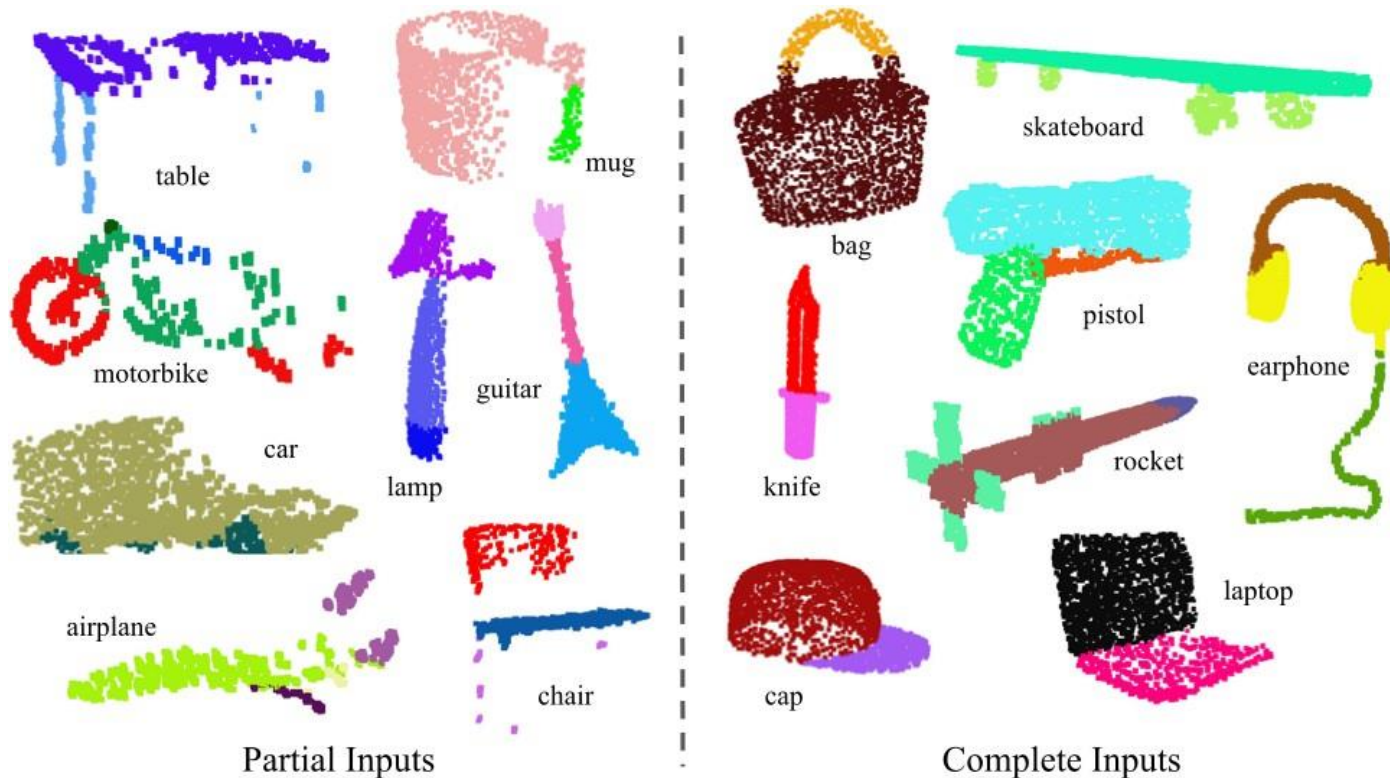
Results on Object Classification

3D CNNs

	input	#views	accuracy avg. class	accuracy overall
SPH [12]	mesh	-	68.2	
3DShapeNets [29]	volume	1	77.3	84.7
VoxNet [18]	volume	12	83.0	85.9
Subvolume [19]	volume	20	86.0	89.2
LFD [29]	image	10	75.5	-
MVCNN [24]	image	80	90.1	-
Ours baseline	point	-	72.6	77.4
Ours PointNet	point	1	86.2	89.2

dataset: ModelNet40; metric: 40-class classification accuracy (%)

Results on Object Part Segmentation



Results on Object Part Segmentation

	mean	aero	bag	cap	car	chair	ear phone	guitar	knife	lamp	laptop	motor	mug	pistol	rocket	skate board	table
# shapes		2690	76	55	898	3758	69	787	392	1547	451	202	184	283	66	152	5271
Wu [28]	-	63.2	-	-	-	73.5	-	-	-	74.4	-	-	-	-	-	-	74.8
Yi [30]	81.4	81.0	78.4	77.7	75.7	87.6	61.9	92.0	85.4	82.5	95.7	70.6	91.9	85.9	53.1	69.8	75.3
3DCNN	79.4	75.1	72.8	73.3	70.0	87.2	63.5	88.4	79.6	74.4	93.9	58.7	91.8	76.4	51.2	65.3	77.1
Ours	83.7	83.4	78.7	82.5	74.9	89.6	73.0	91.5	85.9	80.8	95.3	65.2	93.0	81.2	57.9	72.8	80.6

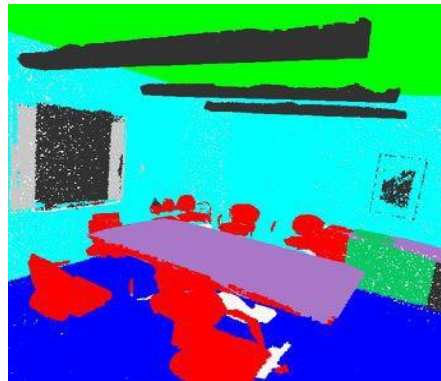
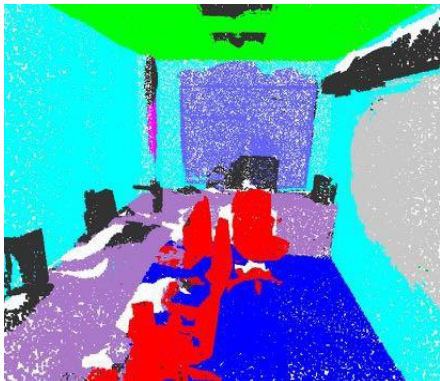
dataset: ShapeNetPart; metric: mean IoU (%)

Results on Object Part Segmentation

Input

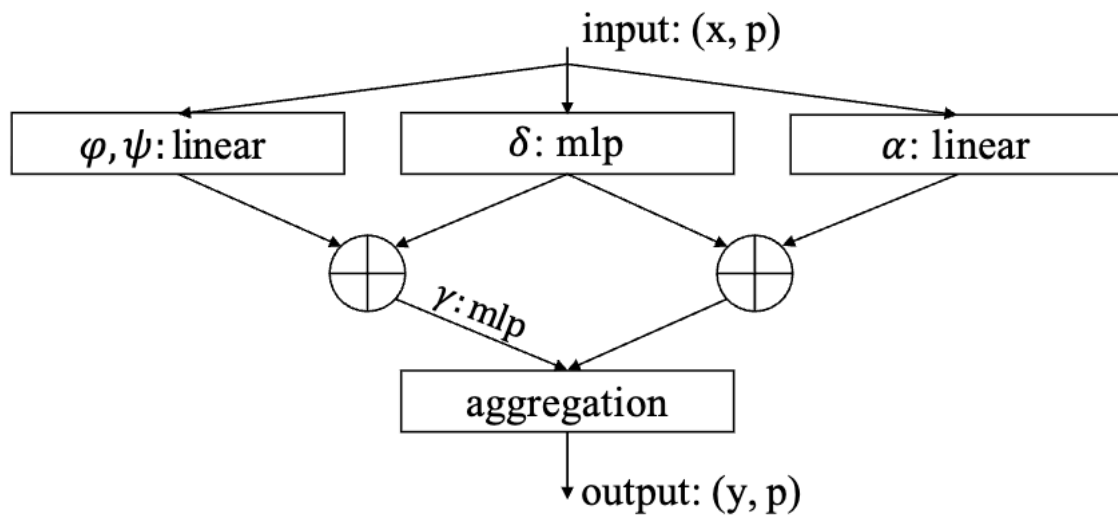


Output



dataset: Stanford 2D-3D-S (Matterport scans)

Point Transformer: The Basic Unit



Point Transformer: The Basic Unit

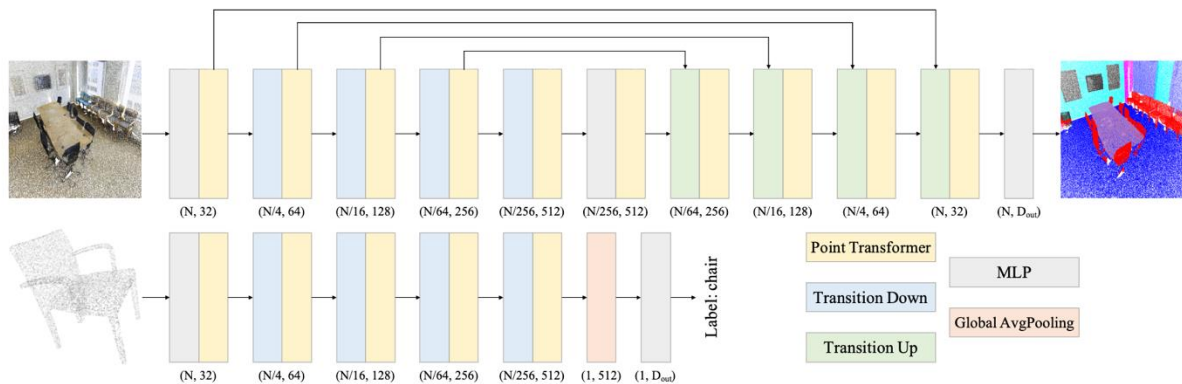
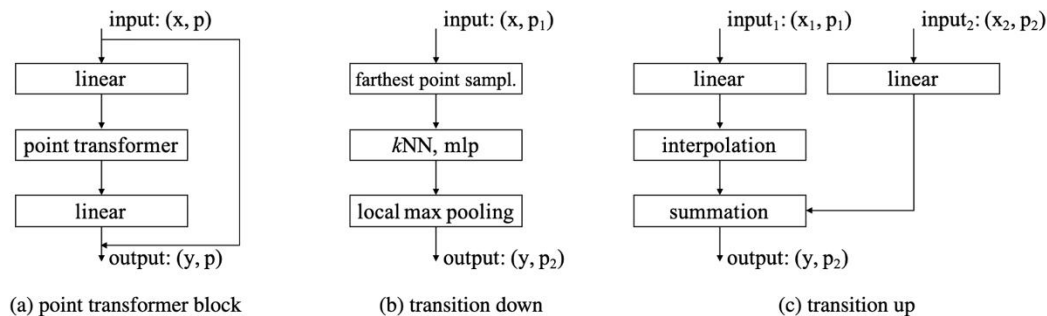


Figure 3. Point transformer networks for semantic segmentation (top) and classification (bottom).

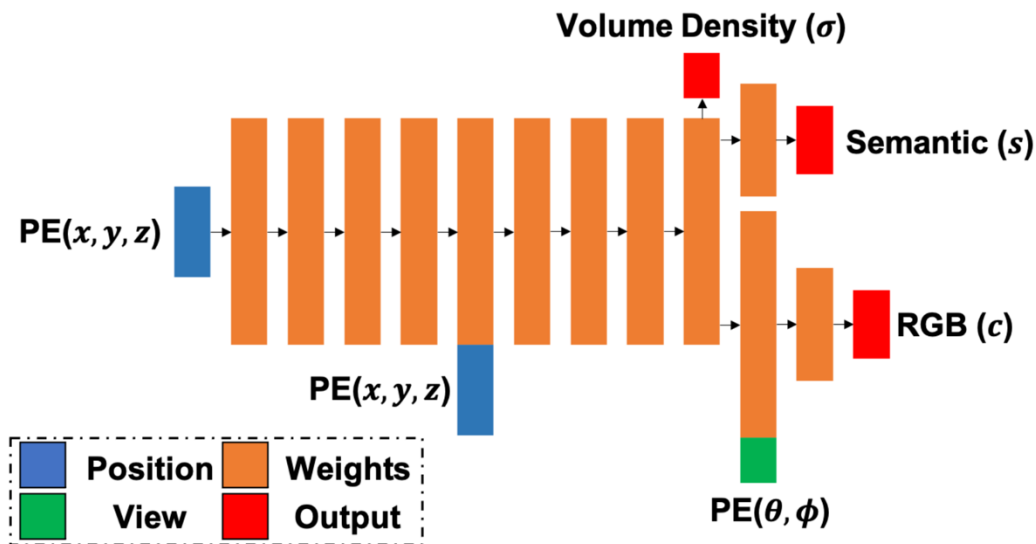


Point Transformer: the Results

Method	OA	mAcc	mIoU	ceiling	floor	wall	beam	column	window	door	table	chair	sofa	bookcase	board	clutter
PointNet [25]	–	49.0	41.1	88.8	97.3	69.8	0.1	3.9	46.3	10.8	59.0	52.6	5.9	40.3	26.4	33.2
SegCloud [36]	–	57.4	48.9	90.1	96.1	69.9	0.0	18.4	38.4	23.1	70.4	75.9	40.9	58.4	13.0	41.6
TangentConv [35]	–	62.2	52.6	90.5	97.7	74.0	0.0	20.7	39.0	31.3	77.5	69.4	57.3	38.5	48.8	39.8
PointCNN [20]	85.9	63.9	57.3	92.3	98.2	79.4	0.0	17.6	22.8	62.1	74.4	80.6	31.7	66.7	62.1	56.7
SPGraph [15]	86.4	66.5	58.0	89.4	96.9	78.1	0.0	42.8	48.9	61.6	84.7	75.4	69.8	52.6	2.1	52.2
PCCN [42]	–	67.0	58.3	92.3	96.2	75.9	0.3	6.0	69.5	63.5	66.9	65.6	47.3	68.9	59.1	46.2
PAT [50]	–	70.8	60.1	93.0	98.5	72.3	1.0	41.5	85.1	38.2	57.7	83.6	48.1	67.0	61.3	33.6
PointWeb [55]	87.0	66.6	60.3	92.0	98.5	79.4	0.0	21.1	59.7	34.8	76.3	88.3	46.9	69.3	64.9	52.5
HPEIN [13]	87.2	68.3	61.9	91.5	98.2	81.4	0.0	23.3	65.3	40.0	75.5	87.7	58.5	67.8	65.6	49.4
MinkowskiNet [37]	–	71.7	65.4	91.8	98.7	86.2	0.0	34.1	48.9	62.4	81.6	89.8	47.2	74.9	74.4	58.6
KPConv [37]	–	72.8	67.1	92.8	97.3	82.4	0.0	23.9	58.0	69.0	81.5	91.0	75.4	75.3	66.7	58.9
PointTransformer	90.8	76.5	70.4	94.0	98.5	86.3	0.0	38.0	63.4	74.3	89.1	82.4	74.3	80.2	76.0	59.3

Table 1. Semantic segmentation results on the S3DIS dataset, evaluated on Area 5.

NeRF with Semantic Segmentation



$$\hat{\mathbf{C}}(\mathbf{r}) = \sum_{k=1}^K \hat{T}(t_k) \alpha(\sigma(t_k) \delta_k) \mathbf{c}(t_k),$$

where $\hat{T}(t_k) = \exp \left(- \sum_{k'=1}^{k-1} \sigma(t_k) \delta_k \right),$

$$\hat{\mathbf{S}}(\mathbf{r}) = \sum_{k=1}^K \hat{T}(t_k) \alpha(\sigma(t_k) \delta_k) \mathbf{s}(t_k),$$

where $\hat{T}(t_k) = \exp \left(- \sum_{k'=1}^{k-1} \sigma(t_k) \delta_k \right),$

NeRF with Semantic Segmentation

Ground Truth



Rendering

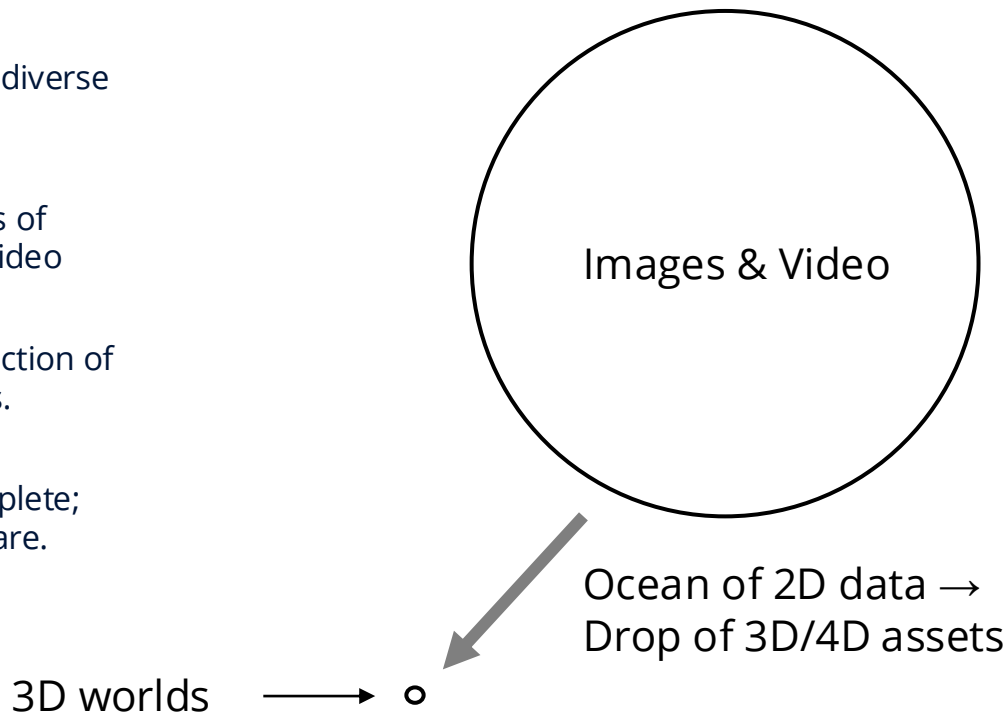


Entropy

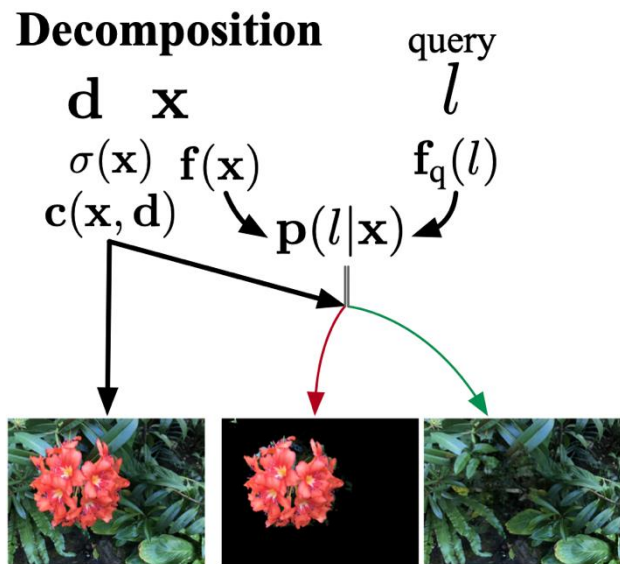
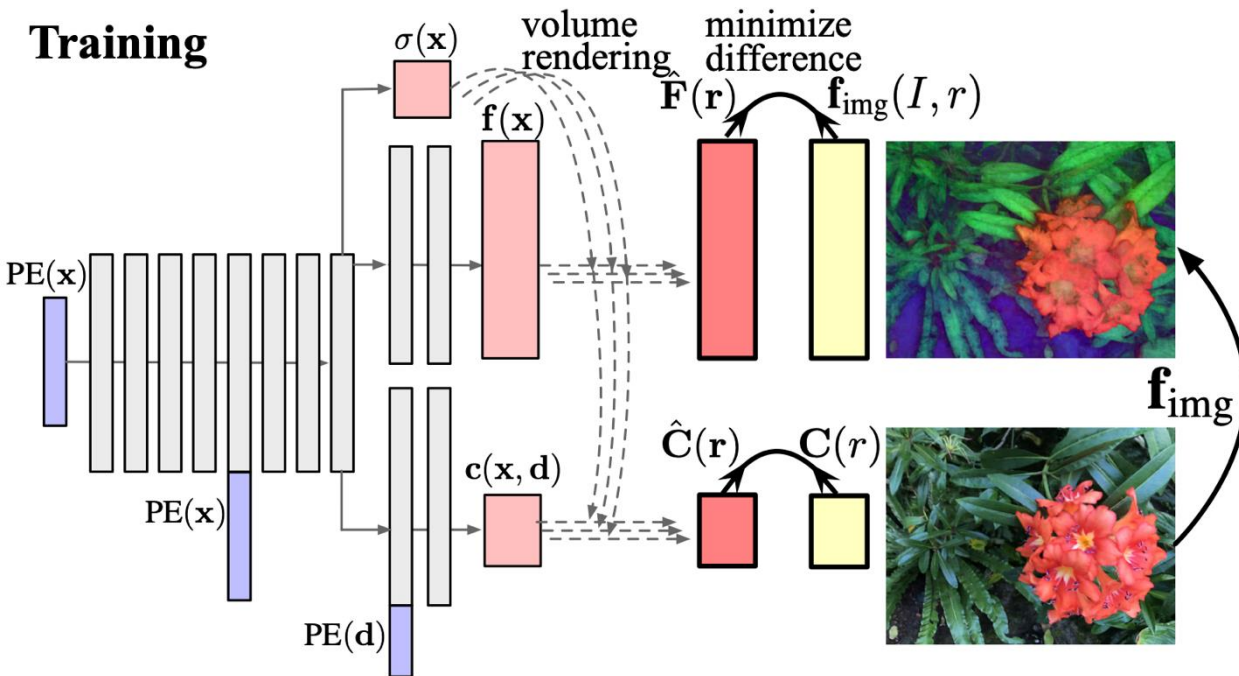


3D Data Are Rare, Image Data Are Everywhere

- Foundation model thrives on large, diverse datasets.
- Images/Video are abundant: billions of users upload roughly a century of video every day.
- 3D worlds are scarce: only a tiny fraction of scenes have high quality 3D models.
- Depth sensors are noisy and incomplete; scene annotations are even more rare.



Can We Achieve 3D Segmentation without Labels?



Can We Achieve 3D Segmentation without Labels?

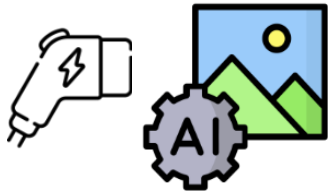


NeRF is Slow, But We Can Accelerate This with 3DGS

In this work, we supercharge the 3D Gaussian Splatting to learn feature fields via knowledge distillation from 2D foundation models.



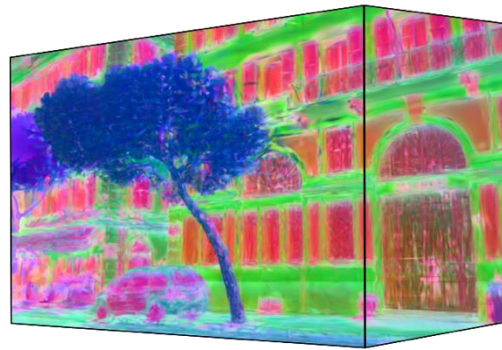
3D Gaussians



2D Foundation Model

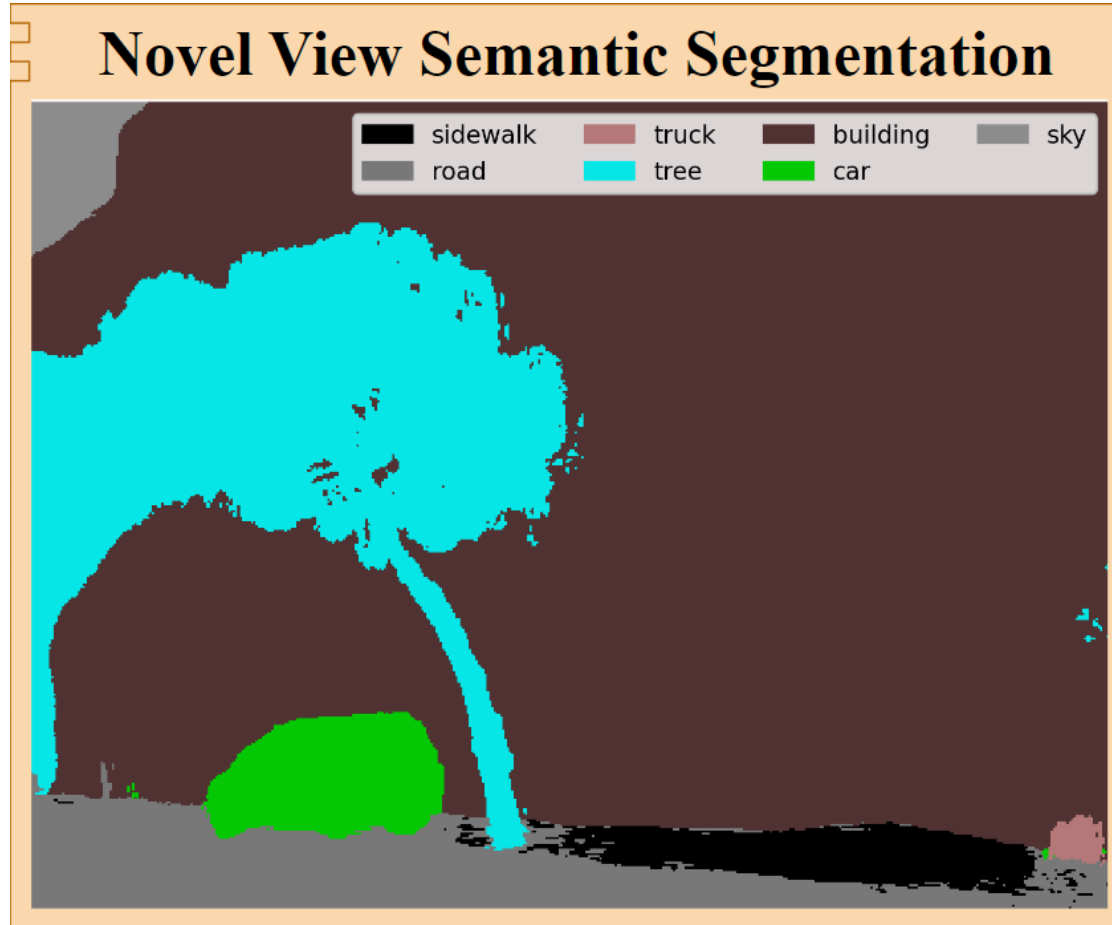


Radiance
Field



Feature
Field

Applications



Novel View Semantic Segmentation



Language-guided Editing



Applications

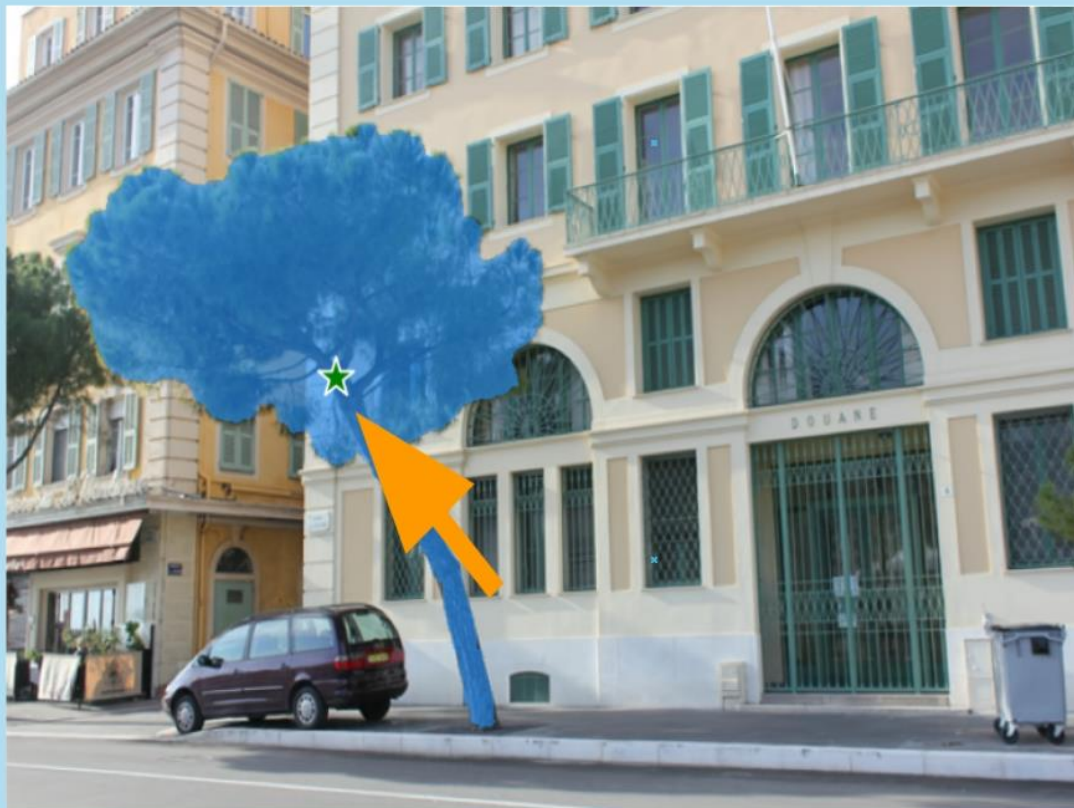
Novel View Semantic Segmentation



Language-guid



Promptable Segment Anything



Applications

Novel View Semantic Segmentation



Promptless Segment Everything



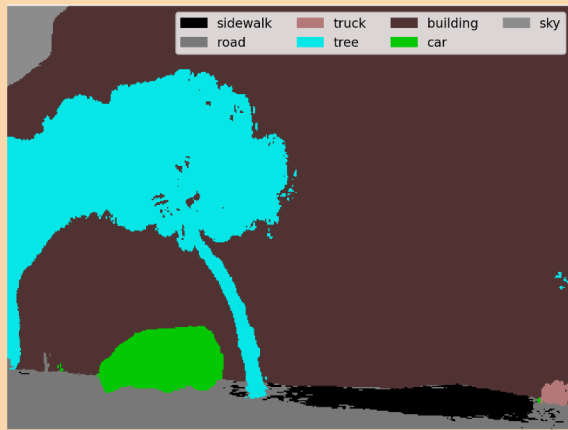
Promptable Segment Anything



Language-guided



Novel View Semantic Segmentation



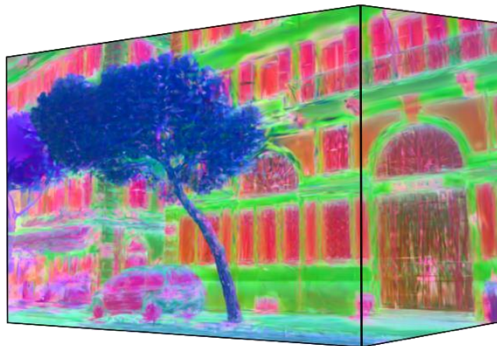
Applications



Promptable Segment Anything

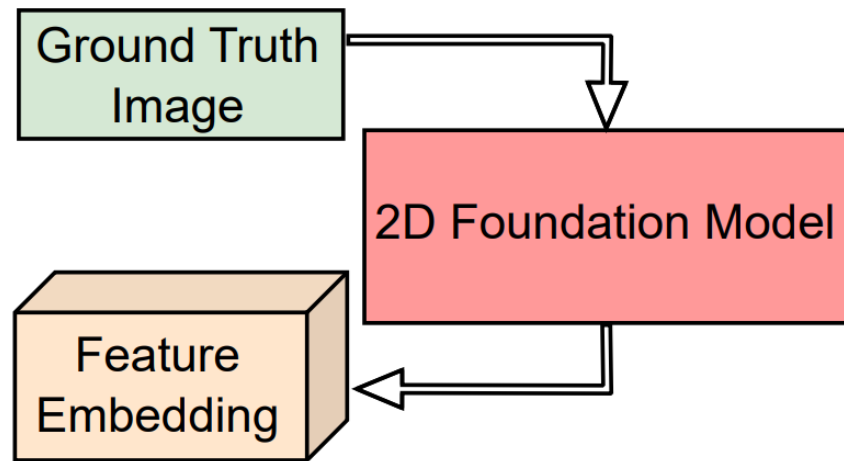
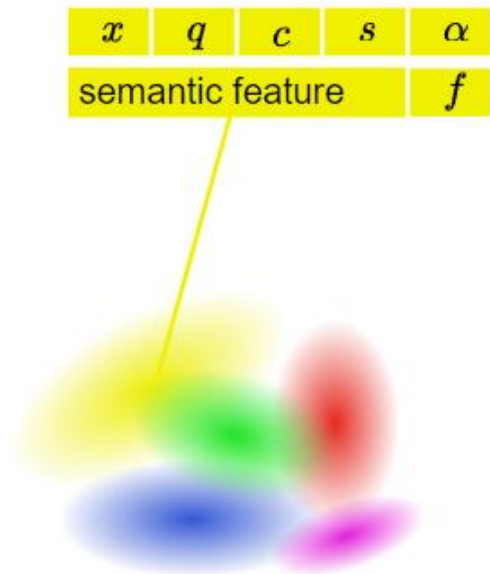


Language-guided Editing

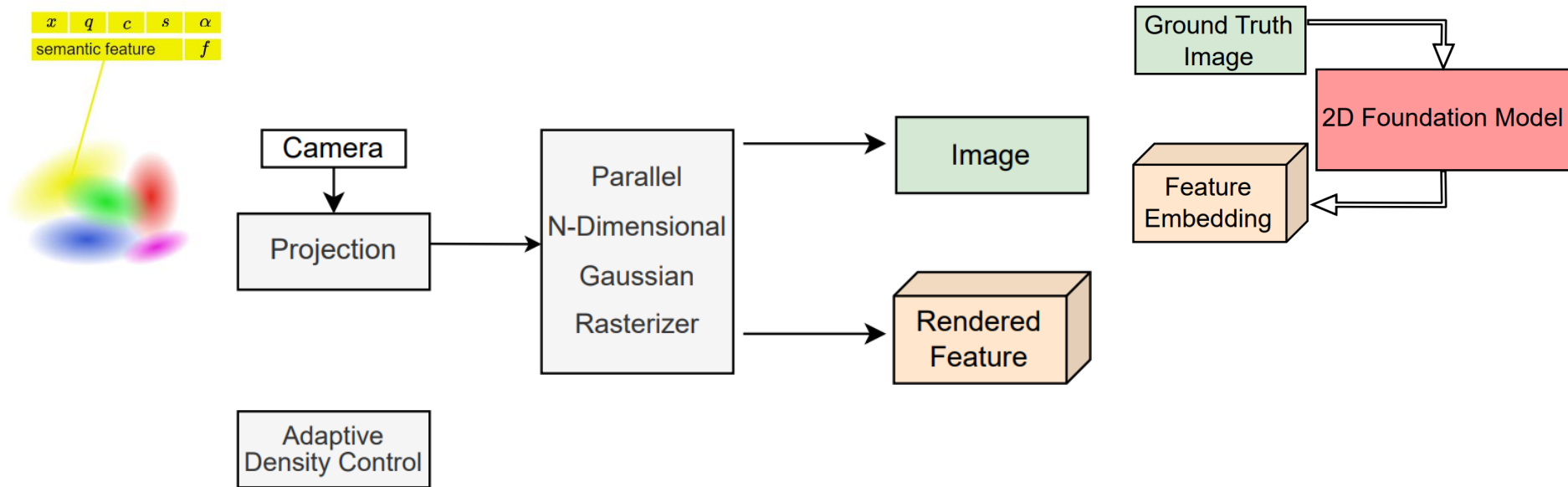


Promptless Segment Everything

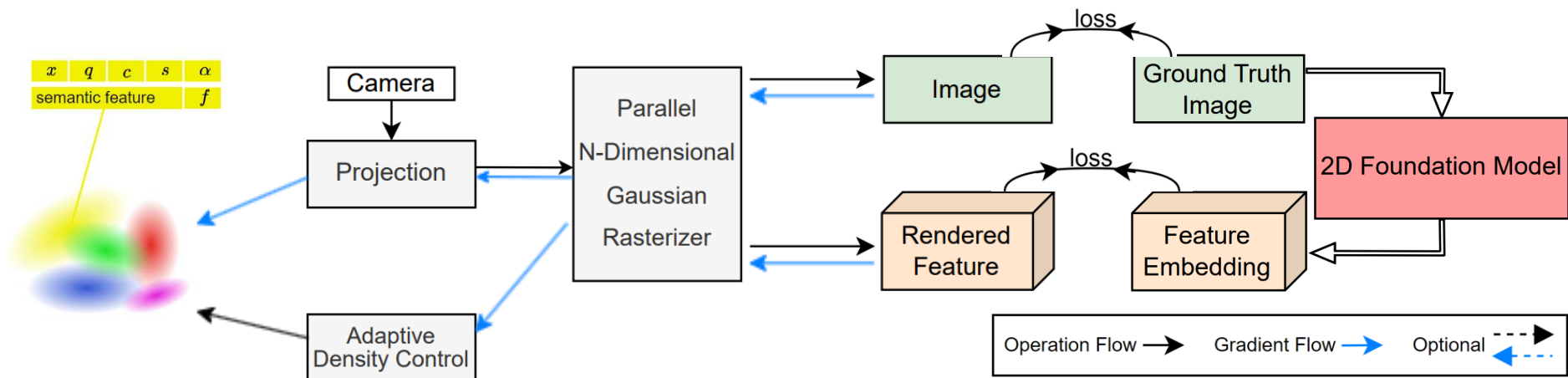




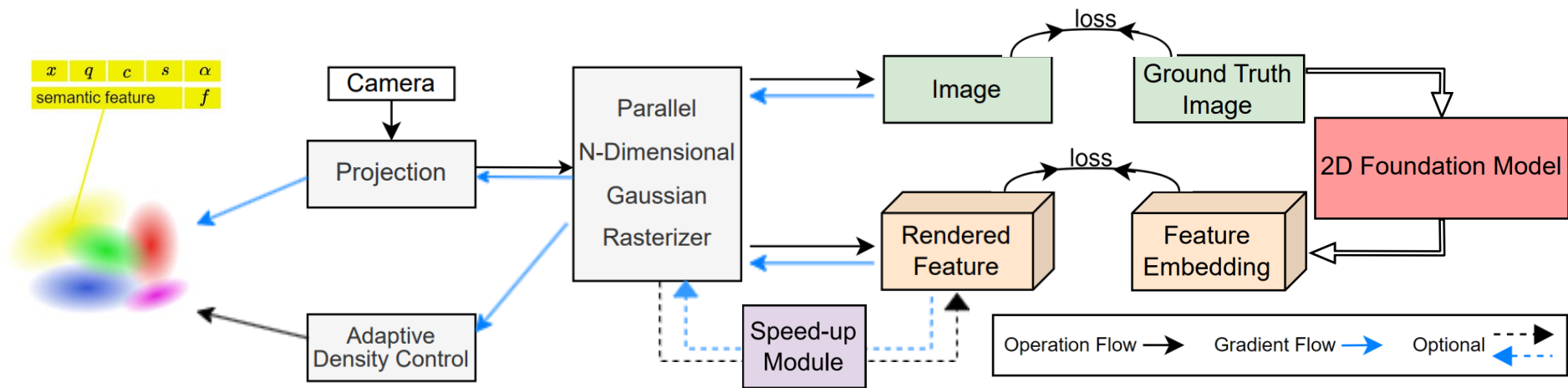
We utilize the state-of-the-art 2D foundation models as our teacher network.



We render the image and feature map via our Parallel N-Dimensional Gaussian Rasterizer.



CUDA empowered computation for end-to-end optimization.



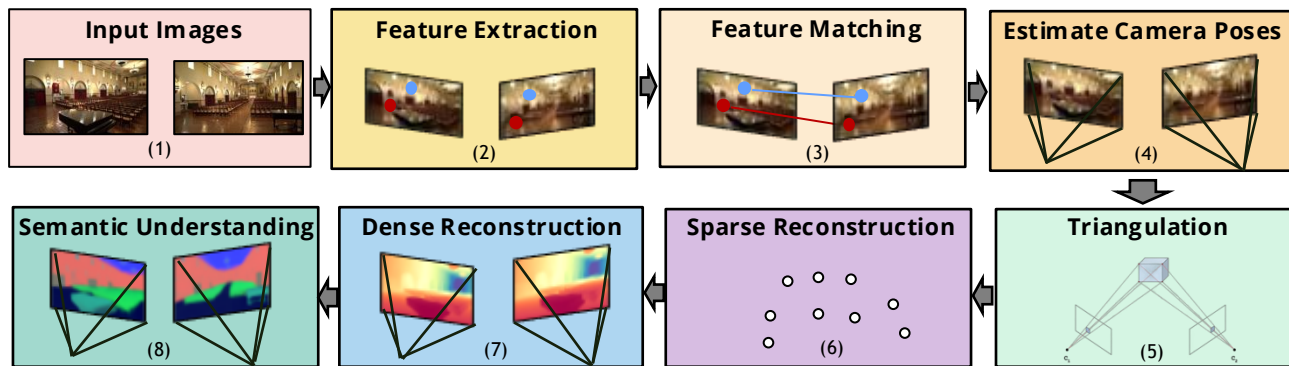
Optional convolutional speed-up module for even faster training and inference speed.

The Full 3D Perception & Reconstruction Pipeline

Raw Images

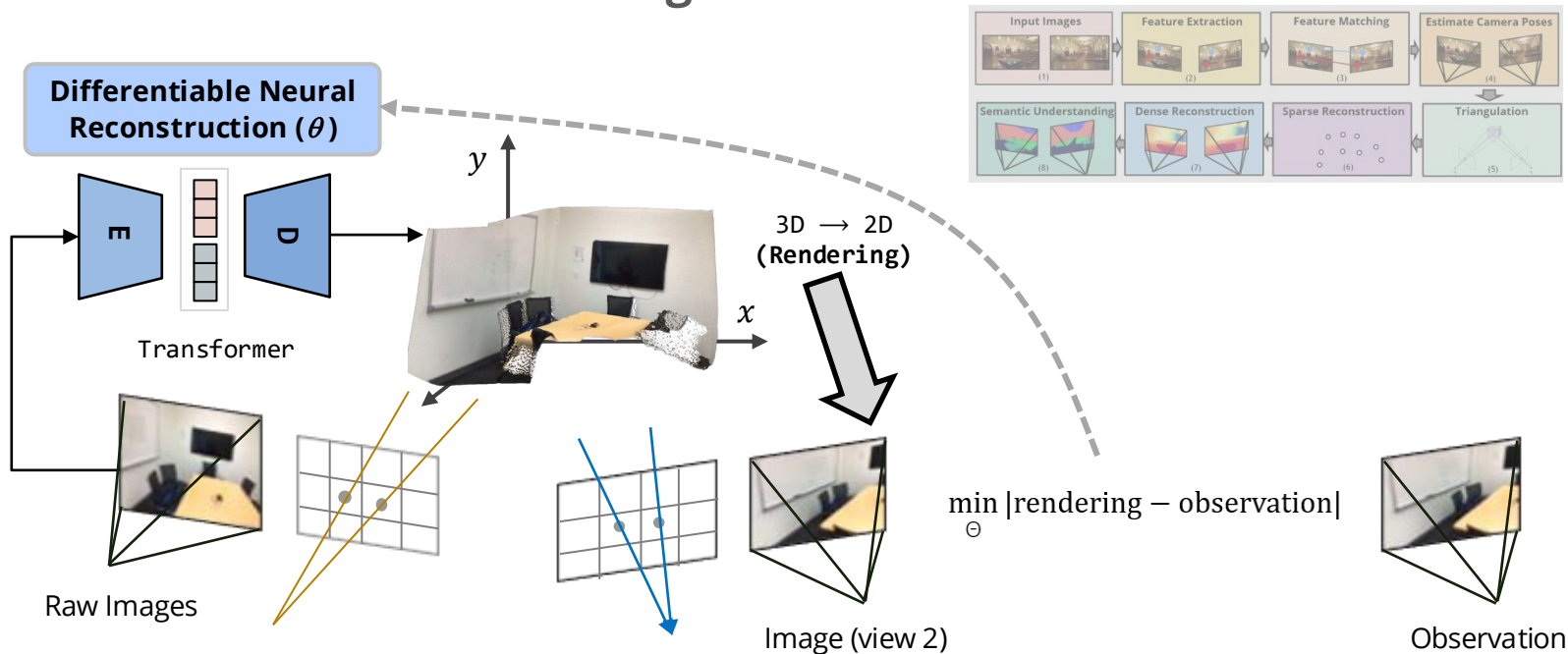


Semantic 3D Maps

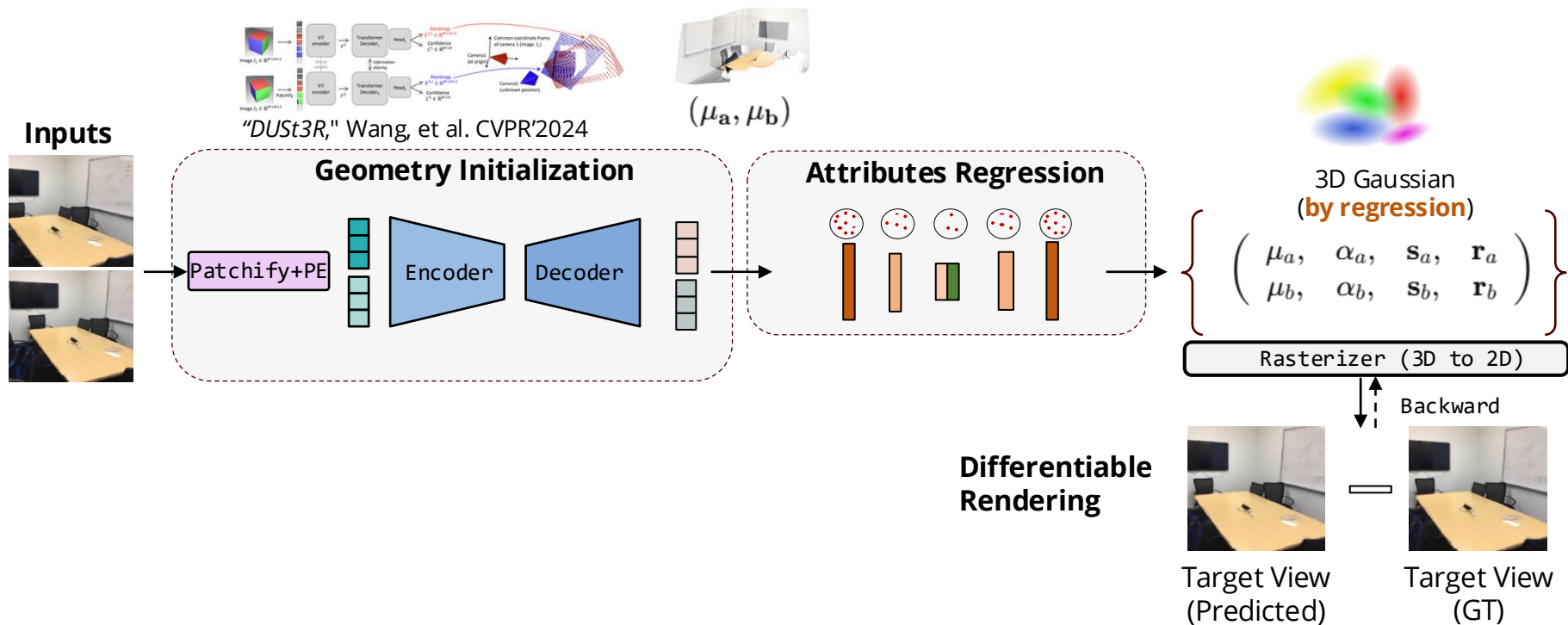


Model Solution: A Fully Differentiable Pipeline

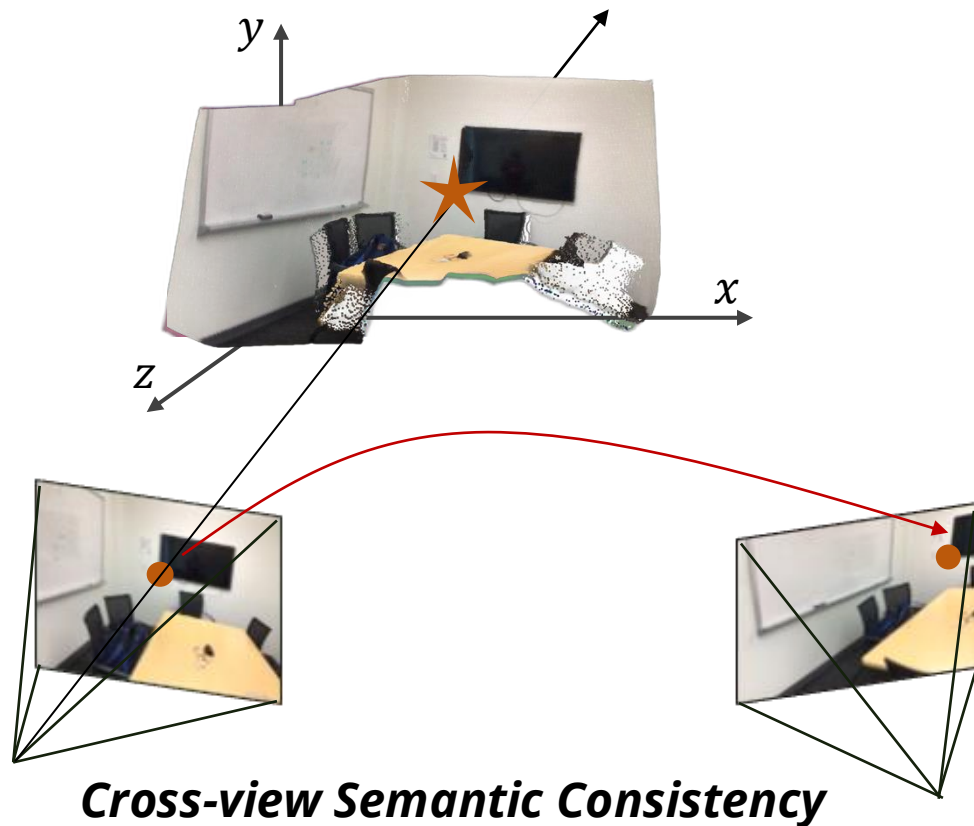
Neural Network + Neural Rendering



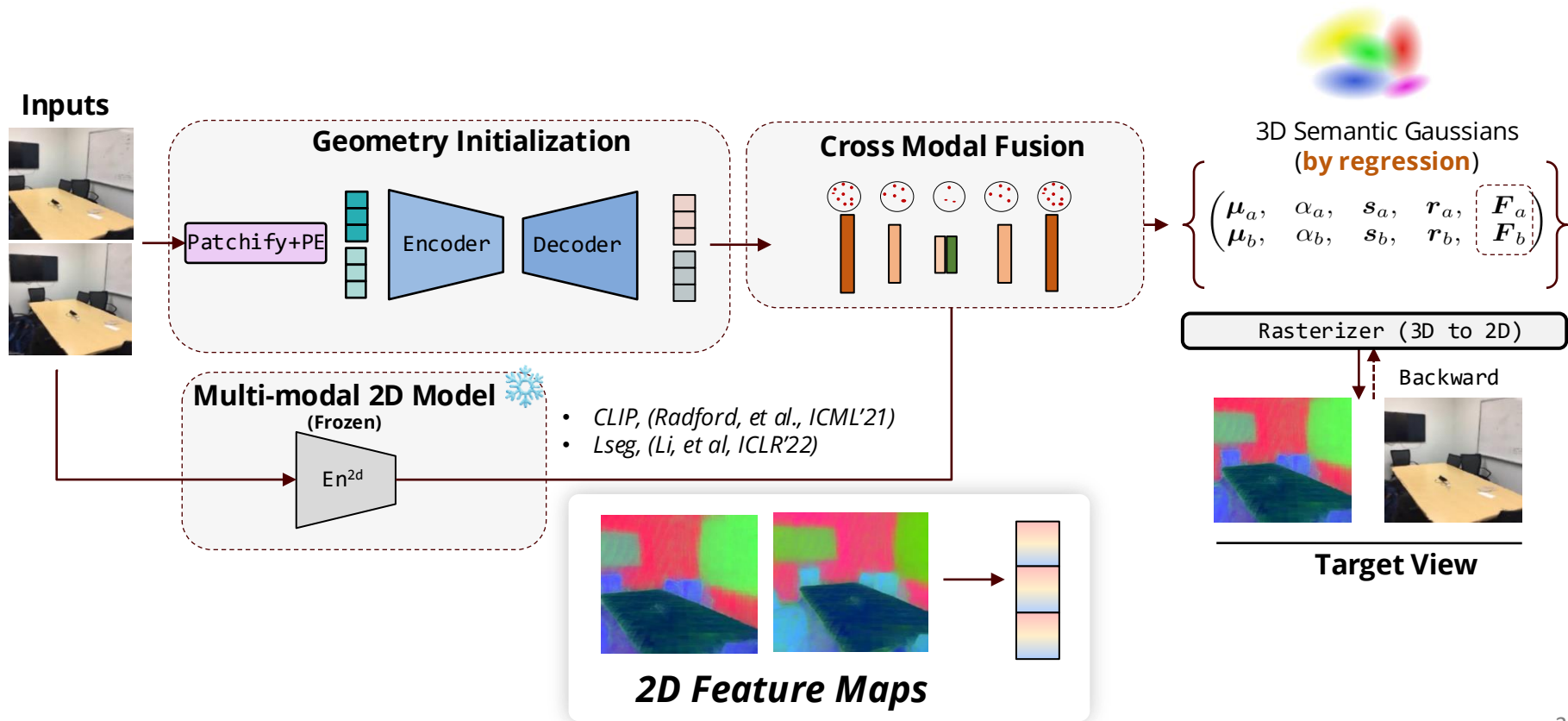
Unified 3D Reconstruction with Gaussian Splatting



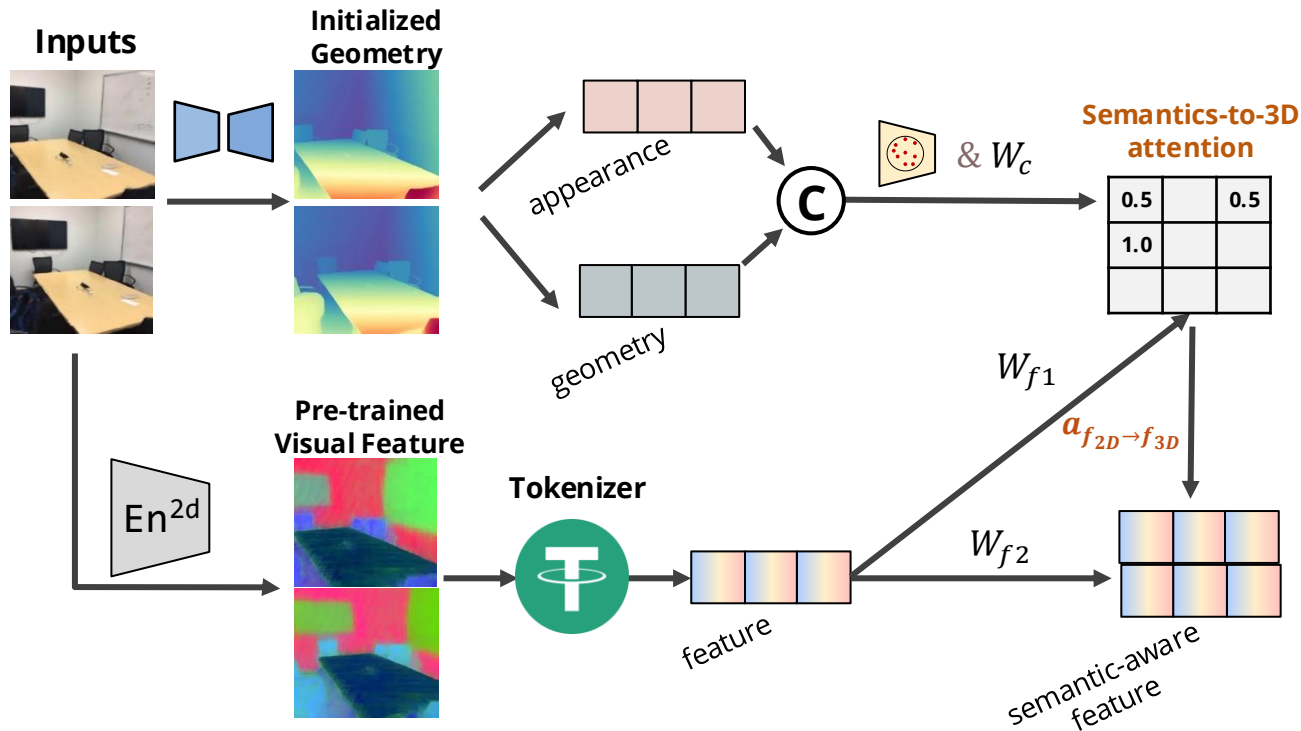
Learning 3D Semantics via Cross-View Consistency



Cross-Modal Fusion for Semantic 3D Gaussians



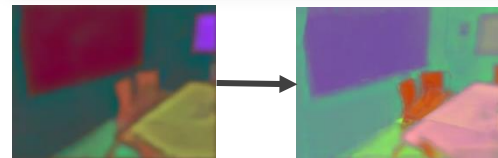
Cross-Modal Attention for Semantic 3D Integration



$$a_{f_{2D} \rightarrow f_{3D}} = \text{softmax} \left(\frac{\text{Proj}(\mu, c) \cdot \text{Proj}(f_{2D})}{\sqrt{d}} \right)$$

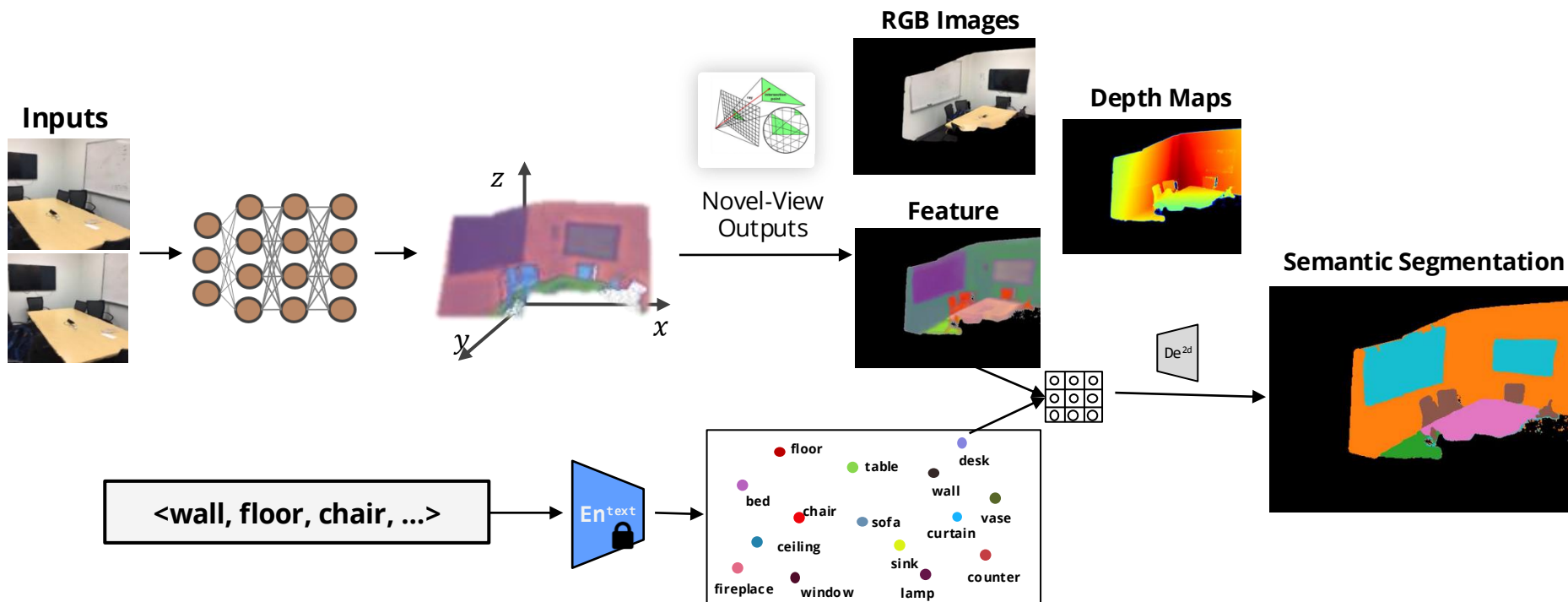
Key idea:

- Per-view 2D features vary
- Attention ensures consistency

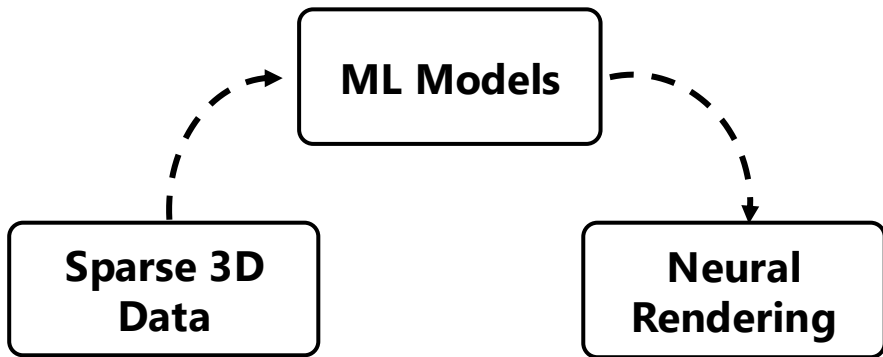


Inconsistent 2D Consistent 3D

Text-Driven Semantic Queries over 3D Maps



Scaling Semantic 3D Mapping: Zero-Shot Generalization



Train one model $\longrightarrow$ **Generalize in a zero-shot setting**